

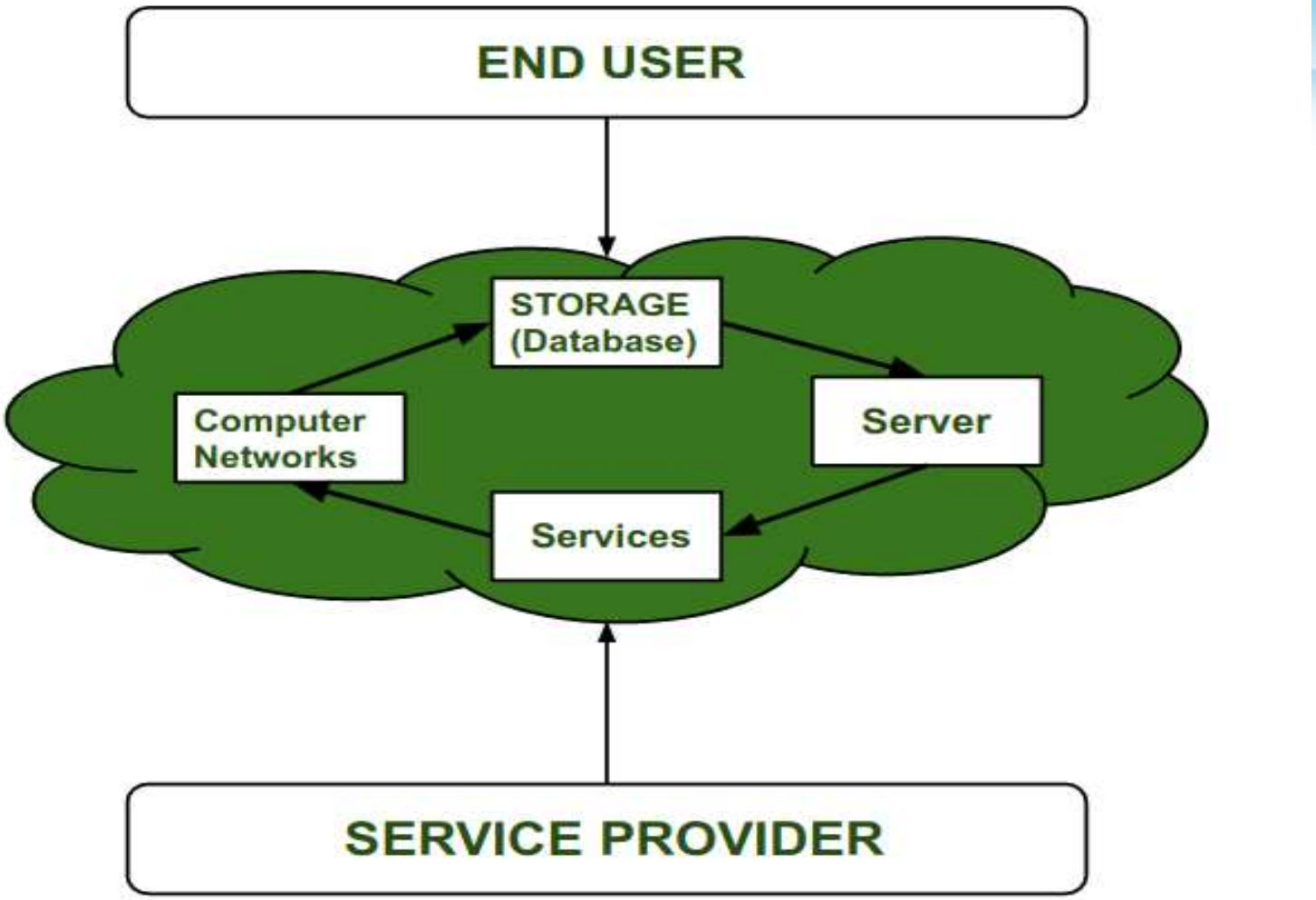
- **Challenges of cloud computing**
- **Security Issues in Cloud**
- **Attacks based on cloud components**
- **Major Public Cloud Cybersecurity Incidents Around the Globe**

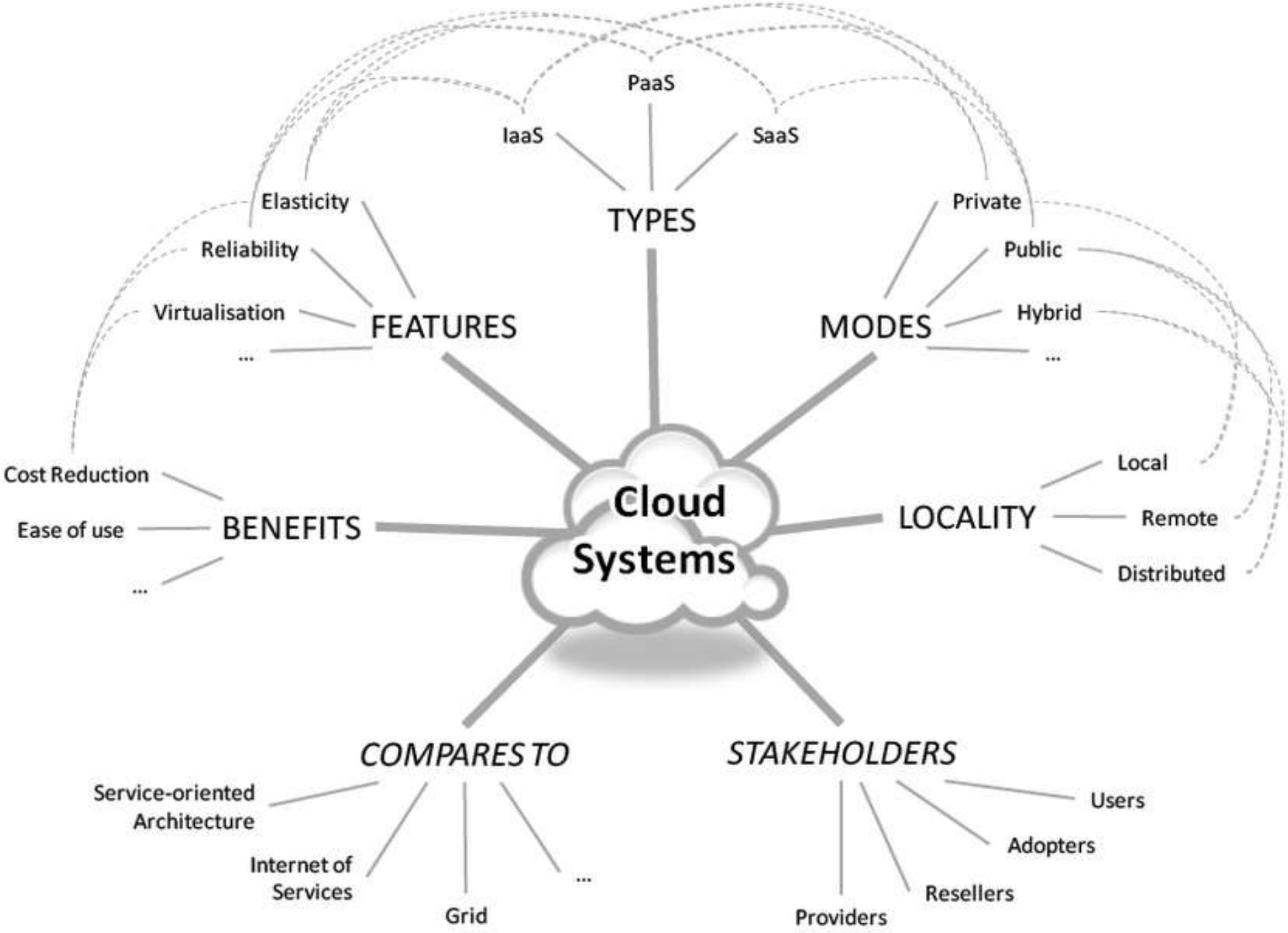
Outline

- **Challenges of cloud computing,**
- **Security issues in cloud,**
- **Categorization of attacks based on cloud components.**



Cloud Computing




Cloud Systems
TYPES
IaaS
PaaS
SaaS
FEATURES
Elasticity
Reliability
Virtualisation
...
MODES
Private
Public
Hybrid
...
BENEFITS
Cost Reduction
Ease of use
...
LOCALITY
Local
Remote
Distributed
COMPARES TO
Service-oriented Architecture
Internet of Services
Grid
...
STAKEHOLDERS
Users
Adopters
Resellers
Providers

Challenges of cloud computing

Cloud security challenges encompass the operational difficulties and complexities that organizations encounter while implementing and managing effective security measures in cloud environments.

Challenges of Cloud Computing

Like any new technology, the adoption of cloud computing is not free from issues.

Some of the most important challenges are as follows:

1. **Security and Privacy**
2. **Interoperability and portability**
3. **Reliability and availability**
4. **Performance and bandwidth cost**
5. **SLA & service quality**
6. **Lack of skill, knowledge and expertise**
7. **Recovery and loss of data**
8. **Insecure APIs**
9. **Vendor Lock-in**



Cloud Computing is **a security nightmare** and it can't be handled in traditional ways.

John Chambers
Former CISCO CEO

Security and Privacy

- How the businesses address the security and privacy concerns thinking of adopting it?
- The valuable enterprise data will reside outside the corporate firewall raises serious concerns.
  - Hacking and various attacks to cloud infrastructure would affect multiple clients even if only one site is attacked.



Interoperability and Portability

- The industry lacks standards for APIs and cloud interfaces, interoperability standards, and associated technical standards that allow interoperability of private to private clouds, public to private clouds, and so on.
- Portability refers to the ability of an application to move across environments, not just across platforms. Portability is a form of reusability.

- Due to lack of portability enables the migration of cloud services from one cloud provider to another or between a public cloud and a private cloud is difficult.



Reliability and availability

- Cloud outages experienced by its providers such as *Google and Amazon* are well documented and publicized.
- This has discouraged potential organizations who were thinking of moving into the cloud environment.
- Cloud computing is a trust-based technology where lack of trust inhibits its adoption.



Performance & bandwidth cost

- Businesses can save money on hardware but they have to spend more for the bandwidth.
- This can be a low cost for smaller applications but can be significant.
- Any enterprises adopting cloud computing services certainly expect the kind of improved performance that an elastic computing environment should provide.

- But cloud services are not perfect. Questions about performance are sure to arise in even the most efficient, well-designed cloud environment.



SLA and Service Quality

- Service quality is also one of the biggest factors considered by enterprises and for this reason, they don't shift their business application to the cloud.
- Though there is an SLA, clients fear that the cloud service providers do not provide any guarantee to ensure product applications security keeps the organizations away from having it.

- QoS approaches in cloud computing have become an important topic in the cloud computing area and there remain open challenges and gaps which require future research exploration.



Lack of skill, Knowledge and expertise

- Every organization does not have sufficient skill or knowledge about the implementation of the cloud solutions.
- organizations have not expertise/skilled staff and tools for the proper use of cloud technology.

- Despite growth in adoption, a cloud skills gap still exists.

- Firms hunt for candidates with more specialized skills related to public cloud and open source platforms.



Recovery of Lost data

- The potential financial impact from losing customer trust is so high on the threats list.
- Cloud services faces issue of data loss.
- A proper backup policy for the recovery of data must be placed to deal with the loss.

  - Vendors must set proper infrastructures to efficiently handle with server breakdown and outages.



Insecure APIs

- Application programming interfaces (APIs) play a significant role in communicating with the cloud services offered by cloud providers.

- APIs define the protocols and methods for requesting and exchanging data, allowing applications to access various cloud resources such as storage, computing power, and databases.

- Insecure APIs can be easily attacked by hackers, and sensitive data can be exposed, leading to data leaks, account takeovers, and service disruptions.

Vendor Lock-In

- The **vendor lock-in** is the situation where customers are dependent on a single **cloud provider** technology implementation and cannot easily move in the future to a different **vendor** without substantial costs, legal constraints, or technical incompatibilities.

- Vendor lock-in is a major barrier to the adoption of cloud computing, due to the lack of standardization.

  - For companies that come to rely heavily on public and hybrid cloud platforms, there is a danger that they become forced to continue with a specific third-party vendor simply to retain operational capacity.



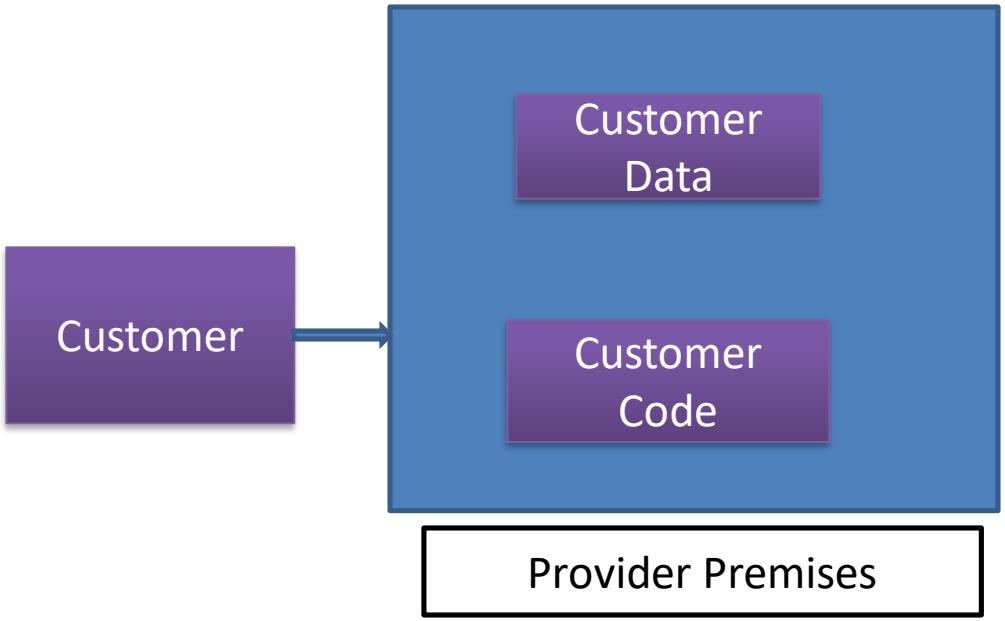
Other issues regarding cloud computing

i) Both customer data and program are residing at cloud provider's premises,
ii) Loss of control,
iii) Lack of trust (Service level agreement), and
iv) Multi-tenancy,
v) Lack of visibility

Cloud Security !! A major Concern

- Security concerns arising because both customer data and program are residing at Provider Premises.

- Security is always a major concern in Open System Architectures (OSA).





Cloud Computing brings new threats

- Traditional system security mostly means keeping bad guys out.
- The attacker needs to either compromise the auth/access control system, or impersonate existing users.

- Cloud Security problems are coming from :
  - **Loss of control**
  - **Lack of trust (mechanisms)**
  - **Multi-tenancy**



Cloud Computing brings new threats

Consumer's loss of control:

- Data, applications, resources are located with provider.
- User identity management is handled by the cloud.
- User access control rules, security policies and enforcement are managed by the cloud provider.
- Consumer relies on provider to ensure-
  - **Data security and privacy**
  - **Resource availability**
  - **Monitoring and repairing of services/resources**



Cloud Computing brings new threats: SLA

- The on-demand service or utility-based economic model necessitates the use of well-established service level agreements (SLAs).
- An SLA is a part of a service contract between the consumer and provider that formally defines the level of service. **It records a common understanding about services, priorities, responsibilities, guarantees and warranties.**

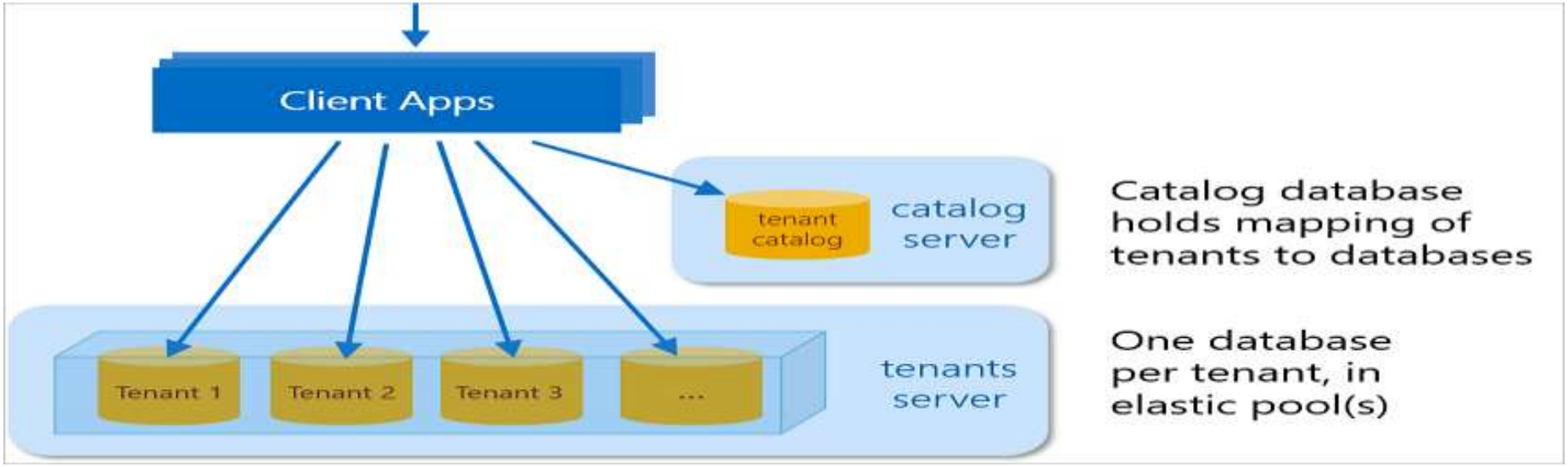
- Therefore, the main issue for cloud computing is to build a ***new layer to support a contract negotiation phase between service providers and consumers and to monitor contract enforcement*** for building trust to the clients.



Cloud Computing brings new threats

Multi-tenancy :

- Multiple independent users share the same physical infrastructure
- So, an attacker can legitimately be in the same physical machine as the target



Lack of visibility

- A typical cloud environment comprises IaaS, SaaS, and PaaS components, often combined with on-premises data centers in hybrid models.

- The rapid proliferation of third-party applications, digital identities, and sensitive data can lead to cloud sprawl, where an organization loses control over its cloud resources.

- The lack of comprehensive visibility slowed down Target's incident response, allowing the breach to persist and cause financial and reputational damage.

Security Issues in Cloud

Security Issues in Cloud

- A security issue is something happening in any assets attacks, misconfiguration, fault, damage, loopholes, and weakness in the system.
- The Cloud Security Alliance's 2026 report identifies 11 leading cloud security threats based on industry surveys.

i) Cloud misconfiguration,
ii) Identity and Access Management (IAM) attacks,
iii) Insecure APIs and interfaces,
iv) Data breaches and unauthorized data access,
v) Account hijacking and credential theft,
vi) Cloud ransomware attacks,
vii) Distributed Denial of Service (DDoS) &
viii) Insider threats.
ix) Insecure virtualization and hypervisor vulnerabilities
x) Supply chain attacks
xi) Advanced Persistent Threats (APTs)



Cloud misconfiguration

High risk

- Misconfiguration is one of the most common causes of cloud security incidents.

Examples:

- Publicly accessible cloud storage buckets.
- Databases exposed to the internet.
- Incorrect firewall and security group rules.
- Unprotected virtual machines.
- Incorrect access permissions.
- Exposed administrative interfaces.
- Insecure default configurations.

Example attack scenario:

- An organization stores confidential research data in cloud storage.
- The storage bucket is accidentally configured for public access.
- An attacker discovers the exposed storage location.
- The attacker downloads confidential files.
- Sensitive information is disclosed without compromising the cloud provider itself.
- Security mechanisms:

Identity and Access Management (IAM) attacks

Critical risk

- Identity and Access Management controls who can access cloud resources and what operations they can perform.

Major vulnerabilities:

- Weak passwords.
- Absence of Multi-Factor Authentication (MFA).
- Stolen credentials.
- Excessive user privileges.
- Poorly managed service accounts.
- Compromised API keys.
- Privilege escalation.
- Session hijacking and token theft.

Example:

- An attacker obtains the credentials of a cloud administrator through phishing. The attacker then logs into the cloud console and accesses virtual machines, databases, and storage systems.

Insecure APIs and interfaces

- Cloud environments depend heavily on APIs for communication between applications, databases, virtual machines, storage, and cloud management services.
- An insecure API can provide attackers with an entry point into the cloud infrastructure.

Common API vulnerabilities:

- Broken authentication.
- Broken authorization.
- API key exposure.
- Excessive data exposure.
- Injection attacks.
- Inadequate input validation.
- Unrestricted API access.
- Rate-limit bypass.

Example attack:

- An attacker discovers an API endpoint that retrieves student information from a university cloud database. Due to inadequate authorization checks, the attacker changes a student ID in the API request and accesses another student's personal information.

Data breaches and unauthorized data access

Critical risk

- A cloud data breach occurs when an unauthorized individual gains access to confidential or sensitive information.

Cloud databases may contain:

- Personal identification information.
- Financial records.
- Medical information.
- Government records.
- University research data.
- Intellectual property.

Authentication credentials.

- Attack mechanisms:
- Stolen credentials.
- SQL injection.
- Misconfigured cloud storage.
- Insider threats.
- Exploitation of vulnerable applications.
- Compromised third-party services.

Account hijacking and credential theft

- Account hijacking occurs when attackers obtain control over legitimate cloud accounts.

Common techniques:

- Phishing.
- Credential stuffing.
- Brute-force attacks.
- Password spraying.
- Session cookie theft.
- OAuth token theft.
- Malware-based credential harvesting.

Consequences:

- Unauthorized access to cloud applications.
- Theft of confidential data.
- Creation of unauthorized virtual machines.
- Modification of security configurations.
- Installation of ransomware.
- Financial losses through unauthorized resource consumption.

Cloud ransomware attacks

- Ransomware is malicious software that encrypts files or otherwise makes organizational data unavailable and demands payment for recovery.

Cloud environments can be affected through:

- Compromised cloud administrator accounts.
- Infected virtual machines.
- Compromised endpoints synchronized with cloud storage.
- Exploited application vulnerabilities.
- Compromised backup accounts.
- Abuse of cloud storage APIs.

Distributed Denial of Service (DDoS)

- DDoS attacks attempt to make cloud-hosted services unavailable by overwhelming them with malicious traffic or requests.
- Types of DDoS attacks:

Attack type	Mechanism
Volumetric	Flooding networks with massive traffic
Protocol-based	Exploiting network protocol weaknesses
Application-layer	Overloading HTTP, DNS or application services
DNS amplification	Using DNS infrastructure to amplify traffic
HTTP flood	Sending excessive HTTP requests

Insider threats

- Insider threats originate from individuals who already have legitimate access to cloud systems.

These may include:

- Employees.
- System administrators.
- Database administrators.
- Developers.
- Contractors.
- Third-party service personnel.

Types:

- Malicious insider: deliberately steals or destroys information.
- Negligent insider: accidentally exposes confidential data.
- Compromised insider: an employee's account is taken over by an attacker.

- Example: A cloud administrator downloads confidential research information and transfers it to an external storage account.

Insecure virtualization and hypervisor vulnerabilities

- Virtualization is fundamental to cloud computing. Multiple virtual machines may operate on the same physical server.
- The hypervisor manages the underlying physical resources and isolates virtual machines.
- A vulnerability in the hypervisor may threaten the isolation between virtual machines.

Major threats:

- Hypervisor exploitation.
- VM escape.
- VM-to-VM attacks.
- Insecure VM images.
- Virtual machine sprawl.
- Snapshot theft.
- Unauthorized VM cloning.
- Side-channel attacks.

Supply chain attacks

- Cloud services depend on many third-party components, including software libraries, APIs, plugins, development tools, managed services, and external vendors.
- An attacker may compromise a third-party component and use it as an entry point into an organization's cloud environment.

Examples:

- Malicious software packages.
- Compromised CI/CD pipelines.
- Vulnerable third-party libraries.
- Compromised software updates.
- Exposed vendor credentials.
- Insecure managed service integrations.

Advanced Persistent Threats (APTs)

- APTs are sophisticated attacks in which threat actors maintain unauthorized access to an organization's systems for an extended period.
- Cloud infrastructure is an attractive target for espionage, intellectual property theft, and strategic information collection.

Typical attack activities:

- Initial compromise through phishing.
- Credential theft.
- Privilege escalation.
- Lateral movement.
- Persistence through cloud identities.
- Data exfiltration.
- Concealment of malicious activities.

Summary of cloud cybersecurity threats

No.	Cybersecurity issue	Major security solution
1	Cloud misconfiguration	CSPM
2	IAM attacks	MFA, RBAC, ZTA
3	Insecure APIs	API Gateway, WAF
4	Data breaches	Encryption, DLP
5	Account hijacking	MFA, UEBA
6	Ransomware	EDR, immutable backup
7	DDoS	CDN, DDoS protection
8	Insider threats	PAM, UEBA
9	Hypervisor attacks	Hypervisor hardening
10	Supply chain attacks	SBOM, SCA
11	APTs	SIEM, XDR, threat intelligence

Attacks based on cloud components

(Network, Virtual Machines storage and applications)

Attacks based on cloud components

- For performing comparative analysis, attacks on a cloud platform based on its components:

1. **Network (A1),**
2. **Virtual machines (A2)**
3. **Storage (A3) and**
4. **Applications (A4)**



Network based attacks

- An intruder may attack a cloud system through its network which may in turn deteriorate
  the quality of cloud services and may even put data privacy/confidentiality at risk.

1. **Port scanning**
2. **Botnets**
3. **Spoofing attacks**

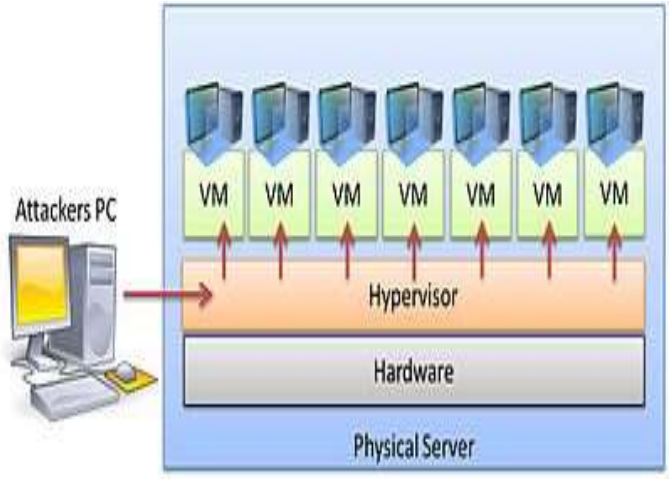




Virtual machine based attacks

- On a cloud system, the VM based attacks exploit vulnerabilities in the virtual machines to violate data protection and affect the cloud services.
- Multiple virtual machines being hosted on a system cause several security risks.

1. **Cross VM side channel attacks**
2. **VM creation attacks**
3. **VM migration and rollback attacks**
4. **VM scheduler based attacks**



Storage based Attacks

- An attacker from outside or even a malicious insider may steal private data stored on some storage device.
- With access to sensitive information, a large number of vulnerabilities may be exploited by manipulating data if a strict monitoring mechanism is not implemented.

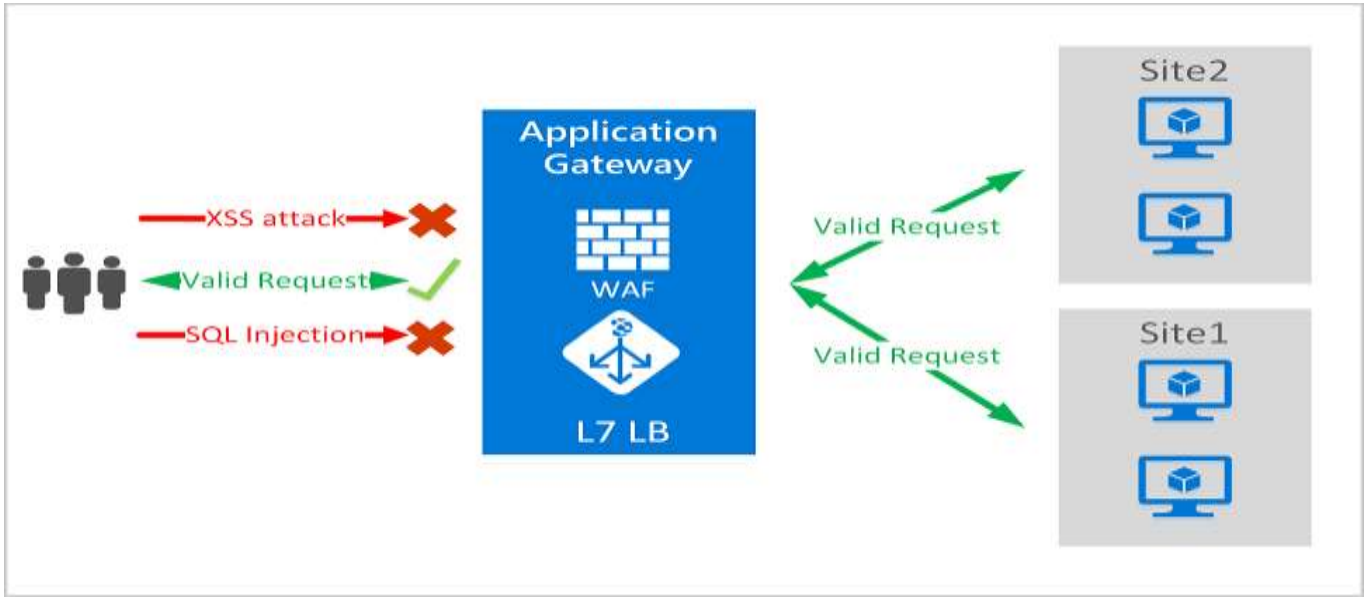
1. **Data scavenging (**the process of searching through a system's data residue to gain unauthorized access to sensitive or valuable information.)

2. **Data reduplication (**a technique that removes duplicate copies of data to improve storage efficiency and reduce the amount of storage media needed.)

Application based attacks

- The applications running on a cloud may be exposed to various attacks by injecting code which may trace execution paths and exploit this information for malicious purposes.

1. **Malware injection attacks**
2. **Shared architectures**
3. **Web services & protocol based attacks**



Major Public Cloud Cybersecurity Incidents Around the Globe

Capital One cloud data breach – USA (2019)

- **Major data breach**
  - Approximately 106 million individuals in the USA and Canada were affected.
  - Customer information was accessed without authorization.
  - The attacker exploited a server-side request forgery (SSRF) vulnerability associated with a web application firewall configuration.
- **Attack mechanism:**
  1. The attacker exploited an application-layer vulnerability.
  2. The vulnerability enabled access to AWS instance metadata.
  3. Temporary AWS credentials were obtained.
  4. Excessive permissions enabled access to cloud storage.
  5. Customer information was downloaded.

Microsoft Azure and Microsoft cloud identity compromise – Global (2023–2024)

Cloud platform: Microsoft cloud identity infrastructure and Exchange Online

- **Identity compromise**
  - The Storm-0558 campaign affected Microsoft customer email accounts.
  - Attackers used forged authentication tokens to access certain Exchange Online accounts.
  - The incident affected US government and other organizational email accounts.

- **Attack mechanism:**
  - Theft of a Microsoft signing key.
  - Forging authentication tokens.
  - Abuse of cloud identity authentication.
  - Unauthorized access to email data.

Snowflake customer data breaches – Global (2024)

- **Cloud platform: Snowflake cloud data platform**
- Large-scale account compromise
  - Multiple customer environments were accessed by attackers.
  - Stolen credentials were used to access customer accounts.
  - Some compromised accounts did not have MFA enabled.
- **Affected organizations included:**
  - Ticketmaster.
  - Santander Bank.
- **Attack mechanism:**
  1. Attackers obtained previously stolen credentials.
  2. They authenticated to customer Snowflake environments.
  3. They accessed and extracted stored information.
  4. Stolen information was used in extortion activities.

Microsoft Midnight Blizzard cloud email compromise – USA (2024)

- **Cloud platform: Microsoft corporate cloud environment**
- **Nation-state intrusion**
  - Microsoft disclosed that the Russian state-sponsored actor Midnight Blizzard had accessed certain corporate email accounts.
  - The attackers used password-spraying techniques against a legacy non-production test tenant.
  - They subsequently exploited an OAuth application to access additional corporate email.
- **Security lessons:**
  - Secure legacy and test environments.
  - Monitor password spraying.
  - Audit OAuth application permissions.
  - Enforce MFA and privileged access controls.
  - Investigate suspicious cloud identity activity.

European Commission AWS cloud breach – Europe (2026)

- **Cloud platform: Amazon Web Services (AWS)**
- Supply-chain compromise, March 2026
  - The European Commission disclosed a compromise affecting its public website hosting platform, europa.eu.
  - Attackers obtained an AWS secret through the Trivy software supply-chain compromise.
  - Approximately 91.7 GB of compressed data was exfiltrated.
  - Personal information and email-related data were among the exposed information.
  - Data associated with other EU entities may also have been affected.
- **Attack mechanism:**
  1. A compromised version of the Trivy security scanning tool entered the software supply chain.
  2. AWS credentials were harvested from the affected environment.
  3. Attackers used the credentials to access an AWS account.
  4. They performed reconnaissance and created additional access credentials.
  5. Data was exfiltrated.

CrowdStrike global IT outage – Worldwide (2024)

- **Technology involved: CrowdStrike Falcon and Microsoft Windows**

- Global operational disruption, 19 July 2024
  - A defective CrowdStrike software update caused Windows systems to crash.
  - Approximately 8.5 million Windows devices were affected.
  - Aviation, banking, healthcare, retail, and other sectors experienced disruptions.

Important distinction:

- **This was not a cloud data breach or a cyberattack. It was a defective security software update with a worldwide operational impact.**
- **It shows the systemic risks of relying on centralized software and cloud-connected security ecosystems.**

Conclusion

- A completely new realm for compute, storage, access, flexibility, productivity and economy of scale has been made possible by the cloud computing environment.

- Cloud Computing also has given rise to further security concerns for the users, IT personnel as well as enterprises or individuals.

- By being aware of cloud computing security challenges and issues, organization's IT team may develop a multi-cloud deployment security strategy to safeguard one's company/enterprise.

Thank You

